

Study 5: Cross-validation

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El Climat – California Housing and decision tree regression

Question

How reliable is one train/test score?



Goal: show why cross-validation gives a more stable estimate of generalization error than training error or one random split.

Task

Predict California district median house value from census and location features.

Model

Decision tree regressor, useful here because it can overfit when unconstrained.

Metric

Mean absolute error (MAE), in hundreds of thousands of dollars.

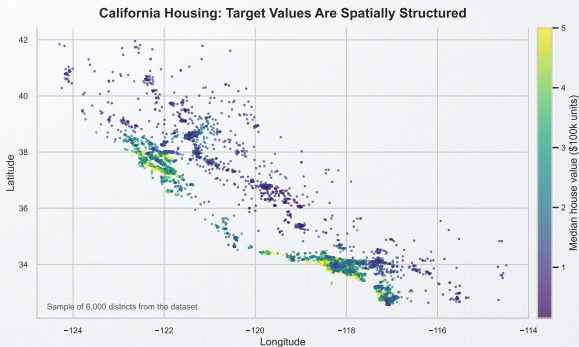
Data

California Housing

A spatial regression problem



SLIDE 4



- 20,640 block groups from the 1990 U.S. census.
- Target: median house value, expressed in units of \$100,000.
- Predictors include income, house age, rooms, population, occupancy, latitude, and longitude.
- The map shows that location carries strong information, so random sampling can matter.

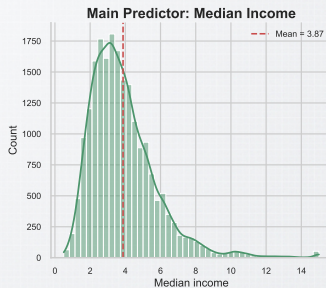
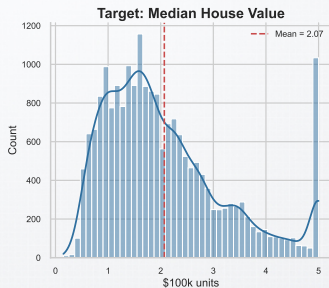
First signals

Income is strong, but not the full story

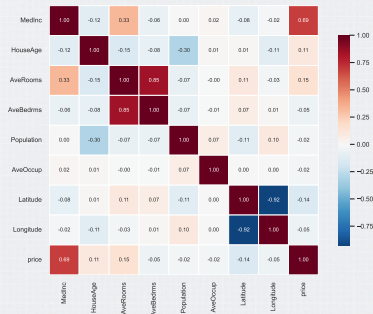


SLIDE 5

The Regression Task: Predict House Value from District Features



Feature Correlations: MedInc Has the Strongest Linear Link to Price

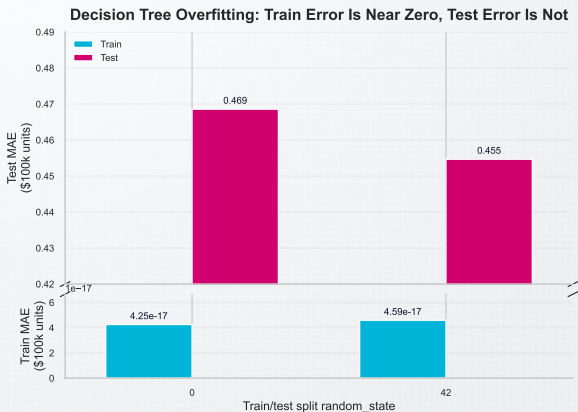


Median income is the strongest linear predictor; geographic variables add spatial structure that a nonlinear model can use.

Evaluation

Training error is misleading

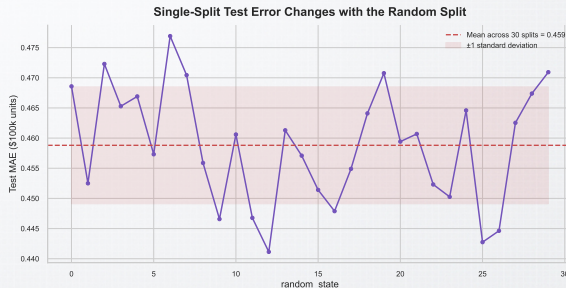
The decision tree memorizes the training data



- The unconstrained tree has almost zero MAE on the data it was trained on.
- Test MAE is much larger, so the near-perfect training score does not describe future performance.
- This is the basic reason evaluation must use unseen data.

One split is sample-dependent

Changing the random split changes the estimate

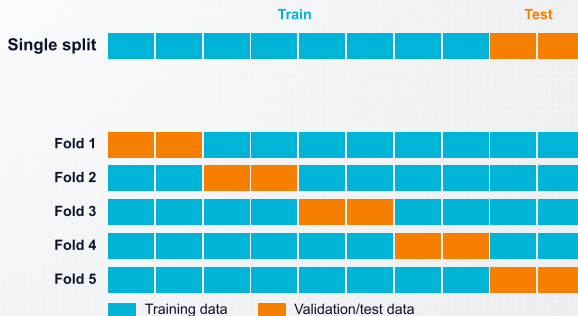


- A train/test split is only one random partition of the same dataset.
- Across 30 random states, the decision tree test MAE moves noticeably.
- A single split is useful, but it can be a noisy estimate of model quality.



Cross-validation

Reuse the data without evaluating on training examples



**Average
fold errors**

MAE = 0.46
Std = 0.01

1. Split the dataset into 20 folds.
2. Train the model 20 times, each time leaving one fold out.
3. Compute MAE on the held-out fold.
4. Summarize the distribution of the 20 errors.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

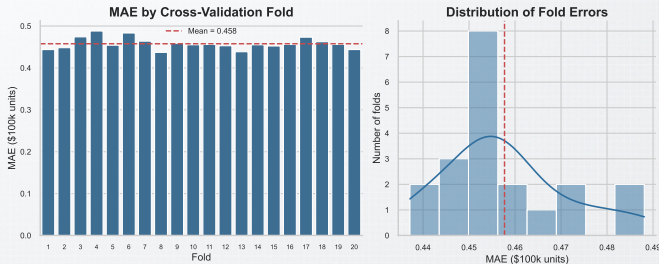
Results

20-fold result

Cross-validation exposes both level and variability



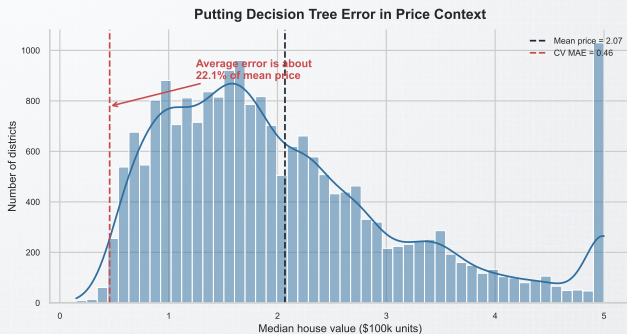
Cross-Validation Gives an Error Distribution, Not Just One Number



- Each bar is a held-out fold, not training performance.
- The distribution gives a central estimate and uncertainty across folds.
- This is more informative than reporting only one random split.

Error scale

The MAE is substantial relative to house prices



- MAE is in the same unit as the target: hundreds of thousands of dollars.
- Comparing MAE with the price distribution makes the number interpretable.
- The tree captures structure, but the average error remains non-negligible.

Consumption forecasting

What changes when the target is energy use?



Temporal signal

Forecasts depend on recent consumption, trend, seasonality, and calendar effects.

External drivers

Weather, holidays, and occupancy can shift the demand profile.

Decision context

Literature stresses evaluation protocol and uncertainty, not only point accuracy.^a

^aTao Hong and Shu Fan.
“Probabilistic electric load forecasting: A tutorial review”.
In: *International Journal of Forecasting* 32.3 (2016),
pp. 914–938. DOI:
10.1016/j.ijforecast.2015.11.011.

Forecast validation

ST4 principle adapted to time series



Avoid leakage

Random K-fold cross-validation can train on future observations and test on the past. That breaks the forecasting setting.

Validate chronologically

Use blocked or rolling-origin validation, for example `TimeSeriesSplit`, then report errors by horizon.^a

^ascikit-learn developers. *TimeSeriesSplit*. 2026.
URL:
https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html
(visited on 06/01/2026).

Conclusion

Takeaways

What Study 5 shows



Study 5 Takeaways

1

Do not evaluate on training data

A decision tree can memorize the training set and report near-zero training MAE.

2

A single test split is only one sample

Changing the random split changes the measured test error.

3

Cross-validation is more informative

20-fold CV estimates mean $\text{MAE} \approx 0.46$ in \$100k units.

Conclusion: cross-validation turns model evaluation from a single fragile score into a distribution of evidence.



Conclusion

Validation is part of the model result

- Evaluating on training data is not a performance estimate; it mainly measures memorization for an unconstrained tree.
- One train/test split is better, but its MAE depends on the random partition.
- Cross-validation turns one fragile number into a distribution of held-out errors.
- For this dataset and model, the cross-validated MAE is large enough to question the unconstrained tree as a final predictor.

Practical rule

Report held-out performance, and when the dataset size allows it, report cross-validation mean and variability rather than only one split.



References

Sources used in the reflection

- [1] Tao Hong and Shu Fan. “Probabilistic electric load forecasting: A tutorial review”. In: *International Journal of Forecasting* 32.3 (2016), pp. 914–938. DOI: 10.1016/j.ijforecast.2015.11.011.
- [2] scikit-learn developers. *TimeSeriesSplit*. 2026. URL: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.TimeSeriesSplit.html (visited on 06/01/2026).

Thank You!

Questions or comments?

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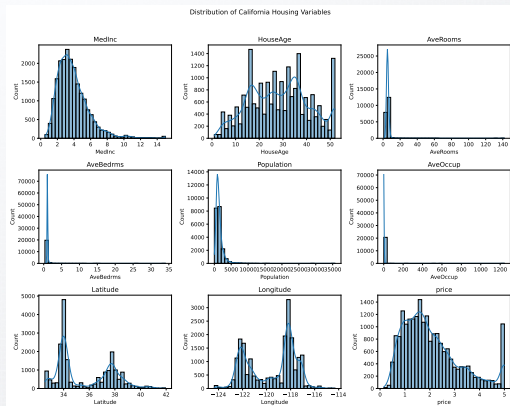
Backup

Raw notebook outputs

Exploratory plots used before polishing



SLIDE 21



- The notebook first checks distributions, missing values, and descriptive statistics.
- These exploratory outputs were simplified into the main presentation figures.
- The full notebook remains available in the project figures folder.

Raw correlation and pairplot

Additional data diagnostics

